UK Environment Data Dashboard Project

Student: Shraddha Yadav | Domain: Environment | Tool: Excel

Dataset source

Source: NESO Regional Carbon Intensity Forecast, National Energy System Operator Data Portal.
URL: https://www.neso.energy/data-portal/regional-carbon-intensity-forecast

The dataset is about regional carbon intensity for the GB electricity system. The measure is gCO2/kWh, which shows how much carbon dioxide is produced for each unit of electricity consumed. I used exactly 3000 records from the CSV file for the Excel dashboard.

1. Dataset selection

I selected an environment dataset because it connects directly with climate and energy use in the United Kingdom. The chosen data is not about one company or one brand. It covers Great Britain electricity regions and shows public information that can help people understand how carbon intensity changes by location and time.

Item	Details
Dataset name	NESO Regional Carbon Intensity Forecast
Domain	Environment - energy and climate
Rows used	3000 records
Geographic scope	Great Britain electricity regions
Main measure	Carbon intensity in gCO2/kWh
File used	regional_carbon_intensity.csv

2. Dataset evaluation and documentation

The data contains half-hourly carbon intensity values for GB electricity regions such as London, South Scotland, East Midlands, Yorkshire and other regions. I prepared the data in Excel by keeping the source, cleaning the time fields, adding date, month, hour and day type columns, and using these fields for analysis tables and charts.

The source is strong because it comes from the National Energy System Operator data portal. It is recent, public, and suitable for a UK public-interest dashboard. The data can show which regions have higher average carbon intensity, which periods have higher values, and how a selected region compares with the GB average.

3. End-consumer identification

Possible users include local councils, sustainability teams, energy analysts, students, and members of the public who want a simple view of electricity carbon intensity. The primary audience chosen

for this dashboard is local council sustainability officers. They need a clear, non-technical dashboard that shows where and when carbon intensity is higher, so they can support energy awareness and low-carbon planning.

4. Perspective and analysis definition

The dataset can be analysed by region, month, hour of day, weekday compared with weekend, and high-carbon events. The chosen perspective is: which GB regions and time periods show higher carbon intensity, and what patterns can be used for low-carbon decision making?

This question matters because carbon intensity is not the same everywhere or at every hour. A simple regional dashboard can help users understand when electricity use is likely to have a higher carbon impact.

5. Dashboard design and development

I redesigned the Excel file so the dashboard is clean and not overcrowded. Instead of placing all charts into one small space, I used three dashboard pages: Overview, Regional Detail, and Time and Distribution. This makes the charts easier to read and reduces label overlap.

Dashboard page	Purpose	Charts included
Dashboard 1 - Overview	High level carbon intensity view	Average by region, monthly trend, hourly trend, high-carbon events
Dashboard 2 - Regional Detail	Regional comparison and selected-region view	Selected region trend, min/avg/max, variability, cleanest regions
Dashboard 3 - Time and Distribution	Time patterns and spread of values	Weekday/weekend, latest interval, distribution bins, London vs GB average

The workbook includes the source sheet first, a 3000-row data sheet, pivot-style analysis tables, dashboard controls, and 12 visible charts. I avoided unsafe manual XML editing because that can make Excel files corrupt. The finished workbook was checked for ZIP integrity and opened/exported through LibreOffice before submission.

Key insights

- Average carbon intensity varies widely across GB regions in the selected records.
- The East Midlands is the highest average region in this sample, while North East England is the lowest average region.
- Hourly patterns show that carbon intensity changes during the day, not just by region.
- The London chart helps compare a major demand area against the GB average.

Files included in submission

The submission folder contains the Excel dashboard workbook, this Word report, and the PDF version of the report. The ZIP file is named ShraddhaYadav_UKDashboard.zip.